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Robust structure-preservation tensorized representation for multi-view unsupervised feature selection

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ABSTRACT

The self-representation methods leverage the linear reconstructability of data points within the same subspace to construct high-quality affinity matrices, showing great potential for improving the performance of multi-view unsupervised feature selection (MUFS). However, existing methods still face some critical challenges. (1) These affinity matrices are inevitably affected by noise in the original data. (2) Enforcing forced non-negativity constraints during their construction may distort the inherent relationships among data points. (3) The optimal structure of the view is not fully exploited. To address these challenges, we propose a novel robust structure-preservation tensorized representation (RSTR) method for MUFS. Specifically, RSTR decomposes each affinity matrix into a clean component and a noise component to mitigate the influence of noise in the original data. Then, the scaled simplex constraint is introduced to ensure the clean affinity matrix more physically meaningful and facilitates the flexible discovery of inherent structures. Moreover, the rank constraint is imposed to identify the optimal structure of each view. Finally, a weighted tensor is constructed from these clean affinity matrices to mine high-order relationships across views. Extensive experiments conducted on eight datasets demonstrate the superiority of RSTR over various state-of-the-art competitors.

1. Introduction

In the real world, data are typically collected by multiple sensors or sourced from multiple origins (Cao & Xie, 2024a; Chen et al., 2022). For example, an essay can be presented in different languages, and an object can be photographed from multiple angles. This type of data is referred to as multi-view data. Compared to single-view data, multi-view data could provide more comprehensive and complementary information. Effectively leveraging such data can significantly enhance the performance of downstream tasks. In recent years, numerous methods have been proposed for classification (Han et al., 2022; Xie et al., 2023) and clustering (Fang et al., 2023b; Guo et al., 2022) tasks on multi-view data, and have achieved remarkable performance. However, the high dimensionality of multi-view data increases computational costs and storage requirements. To mitigate the curse of dimensionality, various dimensionality reduction techniques have been developed, such as feature extraction (Guo et al., 2020; Salau & Jain, 2019) and feature selection (Chandrashekar & Sahin, 2014; Hu et al., 2025). Different from feature extraction techniques, feature selection can perform dimensionality reduction without altering the physical meaning of the features, which could offer better interpretability and garner widespread attention.

According to whether the data have label information, multi-view feature selection can be divided into three categories: supervised (Lin et al., 2022), semisupervised (Van Engelen & Hoos, 2020), and unsupervised (Wang & Song, 2025). Supervised methods typically select the most discriminative features by analyzing the relationships between features and labels. With the guidance of label information, supervised methods can effectively select features that are closely related to the labels, thereby achieving good performance. Semisupervised methods use both the supervisory signals provided by the labeled data and the structural information of the unlabeled data to achieve accurate feature selection. The advantage of these methods is that they can achieve excellent performance even with limited labeled data. Unsupervised methods largely rely on the inherent structure of the data and the relationships among features, which enables them to perform well even in the absence of label information. In practical applications, acquiring labeled data is often expensive and time-consuming. Hence, multi-view unsupervised feature selection (MUFS) has emerged as a significant research direction.

Over the past few years, numerous MUFS methods have been developed (Duan et al., 2025; Fang et al., 2023a; Xu et al., 2025; Yuan et al., 2022; Zhou & Song, 2024). Among them, MUFS methods based

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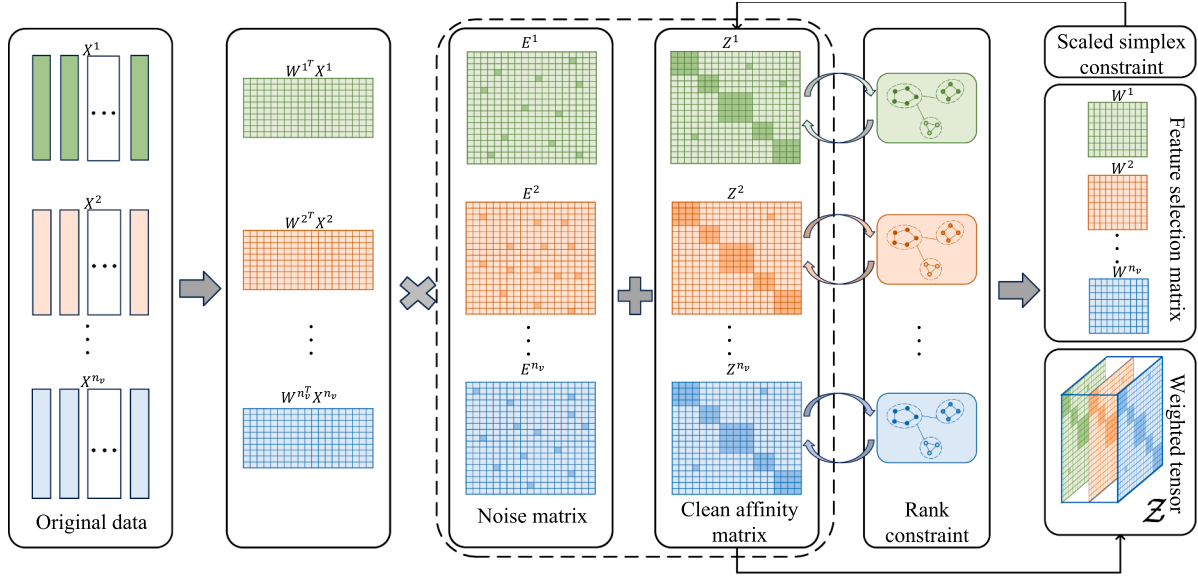


Fig. 1. Framework of RSTR. The original data $\{X^v\}_{v=1}^{n_v}$ are first used to construct affinity matrices, which are then decomposed into noise components $\{E^v\}_{v=1}^{n_v}$ and clean components $\{Z^v\}_{v=1}^{n_v}$. The scaled simplex constraint and rank constraint are then applied to the clean components $\{Z^v\}_{v=1}^{n_v}$ to flexibly capture the inherent relationships within the data points and identify their optimal structures. Subsequently, a weighted tensor $\mathcal{Z}$ is constructed from $\{Z^v\}_{v=1}^{n_v}$ to explore higher-order relationships among different views. Finally, the feature selection matrices $\{W^v\}_{v=1}^{n_v}$ are obtained.

on self-representation have attracted increasing attention. For example, Cao et al. (2024) simultaneously consider the exclusivity of representation and the consistency of indicators to effectively integrate information from multiple views. Cao and Xie (2024b) combine the self-representation with spectral analysis to explore both consistent and inconsistent information across views. Wang et al. (2025a) introduce a graph filtering technique on the original data to denoise and generate smooth representations for each view. Hu et al. (2025) adaptively learns the consistent representation across multiple views and leverages a binary hashing algorithm to guide feature selection. However, MUFS methods based on self-representation still face several key challenges: (1) The affinity matrix is inevitably affected by noise present in the original data. (2) Enforcing non-negativity in the affinity matrix using operations such as absolute value, exponentiation, or squaring might distort the inherent relationships among the data points. In fact, employing a scaled simplex representation is more physically meaningful, as demonstrated in clustering tasks (Cai et al., 2024; Xu et al., 2019). (3) The optimal structure of the affinity matrix is not utilized.

To deal with the above issues, we propose a robust structure-preservation tensorized representation (RSTR) method. The diagram of our method is illustrated in Fig. 1. Specifically, each affinity matrix is decomposed into a clean component and a noise component to reduce the influence of noise in the original data. The scaled simplex constraint is then imposed on the clean affinity matrices to preserve the inherent relationships among data points and enhance the flexibility of the affinity matrices. Meanwhile, the rank constraint is introduced to identify the optimal structure of the clean affinity matrices. Further, a weighted tensor is constructed from these affinity matrices to mine high-order relationships among different views.

The major contributions of this article are distilled as follows:

- We propose a novel MUFS method termed RSTR, which unifies robust self-representation learning, scaled simplex and rank constraints, and tensor learning into a joint framework.
- By introducing an $\ell_{2,1}$ -norm regularization on the noise separated from the self-representation matrices, the influence of perturbations on affinity construction is effectively suppressed.

- The scaled simplex constraint is introduced to enhance the structural flexibility of the affinity matrices, while the rank constraint is used to maintain the optimal structure.
- Extensive experimental results and in-dept analysis demonstrate the effectiveness and superiority of RSTR across eight benchmark datasets compared with state-of-the-art competitors.

The rest of this article is organized as follows. First, Section 2 reviews relevant work. Then, we present the RSTR model and its optimization process in detail in Section 3. We report the experimental results in Section 4 and conclude this article in Section 5.

2. Related work

2.1. Single-view unsupervised feature selection

In recent years, unsupervised feature selection (UFS) has gradually become a prominent research topic due to the widespread availability of high-dimensional unlabeled data. Many studies have shown that feature selection performance can be significantly improved by reducing the impact of noise and outliers in the data (Shang et al., 2023). For example, in Li et al. (2022), Li et al. present a UFS method that combines self-paced learning and subspace learning, aiming to preserve global reconstruction information while reducing the influence of outliers. In Shang et al. (2023), Shang et al. develop a noise-aware UFS method that integrates adaptive graph regularization and self-representation, effectively removing noise from the original data. In Wang et al. (2023), Wang et al. improve the robustness of the model by adding binary weights to directly remove outliers. Feature selection based on pseudo-label learning can generate pseudo-labels for unlabeled data, enabling the exploration of data discriminability in unsupervised scenarios. For instance, in Li et al. (2012), Li et al. propose a UFS method to select the most discriminative features through the joint learning of clustering labels and a feature selection matrix. In Wang et al. (2021), Wang et al. mitigate information loss caused by hard labels by utilizing soft labels to guide feature selection. Preserving the local geometric structure is crucial in enhancing the feature selection performance. In He et al. (2005), He et al. develop a Laplacian score method for computing features through the local geometric structure of the data. In Zhou et al. (2020), Zhou et

al. learn multiple similarity graphs and construct an adaptive consensus graph from these graphs to preserve the local structure among data. In Tang et al. (2023), Tang et al. present a UFS method based on multi-graph fusion and feature weight learning to learn a consensus similarity graph for preserving local structure. In Huang et al. (2023), Huang et al. develop a similarity graph construction strategy based on data relation learning that is used to preserve the local structure of the samples. In recent years, with the diversification of data acquisition methods, traditional single-view UFS methods often struggle to effectively handle multi-view data.

2.2. Multi-view unsupervised feature selection

Multi-view data can provide more comprehensive representations of features by capturing the relationships from diverse perspectives. Thus, MUFS has attracted much attention. In recent years, some MUFS methods have been proposed to mine high-order relationships between data. For example, in Yuan et al. (2022), Yuan et al. present a low-rank tensor constrained graph embedding method to capture higher-order relationships between views. In Song et al. (2024), Song et al. introduce the hypergraphs to capture higher-order relationships between data. In Duan et al. (2025), Duan et al. simultaneously use the hyper-Laplacian regularization and a low-rank tensor to explore higher-order relationships. Exploring the consistency and inconsistency of data is the focus of MUFS. For instance, in Zhou and Song (2024), Zhou et al. propose a multi-view matrix decomposition model to jointly explore both consistency and inconsistency within the data. In Xu et al. (2025), Xu et al. utilize the graph regularization to constrain the consistency matrix, thereby capturing the shared manifold information more precisely. In addition, it employs kernel function to measure the correlation of data in the Hilbert space, effectively integrating the diverse information from all views. Recently, several studies have also taken into account the impact of noise. For example, in Yang et al. (2024), Yang et al. introduce a MUFS method based on self-paced learning, where the model is initially trained using simple samples and then gradually learns complex samples through an adaptive regularizer, effectively avoiding the noise interference. In Wang et al. (2025a), Wang et al. apply the graph filtering to the original data to denoise the data and obtain smooth representations for each view. To capture the global relationships between data, the self-representation learning is widely used in MUFS. For example, in Cao et al. (2024), Cao et al. propose generating multiple graphs that are as mutually exclusive as possible to enhance the complementarity between views, and combine graph learning with consensus clustering to fully leverage the consistency of the indicators. In Wang and Song (2025), Wang et al. present a self-representation-based MUFS method that combines hybrid graph filtering and optimal graph regularization. Although the aforementioned methods have achieved excellent performance, they impose a forced non-negative constraint on the affinity matrix during the self-representation learning process, which may disrupt the internal structure of data.

2.3. Scaled simplex representation

The simplex method is widely used in solving linear programming problems (Patterson & Teh, 2013). Applying simplex constraint to multiple variables (e.g., $x = [x_i]_{i=1}^n$) ensures that they remain non-negative and sum to 1 (i.e., $x^T \mathbf{1} = 1, x_i \geq 0$). It is widely used in clustering due to its simplicity and efficiency (Huang et al., 2015; Nie et al., 2014). To enhance the flexibility of the simplex constraint, in Xu et al. (2019), Xu et al. propose a scaled simplex representation for subspace clustering. Specifically, adjusting the sum of the plurality of variables to a predetermined value s (i.e., $x^T \mathbf{1} = s, x_i \geq 0, 0 < s < 1$), which allows for more flexible utilization of the coefficient vector. In Cai et al. (2024), Cai et al. extend the value of s to $s > 0$, which further enhances the flexibility of the coefficient matrix. Essentially, the simplex constraint can be seen as a special case of the scaled simplex constraint. Most MUFS methods

Table 1

Main notations of our method.

Symbol	Definition
n	The number of samples
d^v	The number of features in the v -th view
n_v	The number of views
c	The number of clusters
$X^v \in \mathbb{R}^{d^v \times n}$	The original data matrix of the v -th view
$W^v \in \mathbb{R}^{d^v \times c}$	The feature selection matrix of the v -th view
$E^v \in \mathbb{R}^{n \times n}$	The noise matrix of the v -th view
$Z^v \in \mathbb{R}^{n \times n}$	The clean affinity matrix of the v -th view
$L_{Z^v} \in \mathbb{R}^{n \times n}$	The Laplacian matrix of the v -th view
$F^v \in \mathbb{R}^{n \times c}$	The indicator matrix corresponding to the Laplacian matrix L_{Z^v}
$I_c \in \mathbb{R}^{c \times c}$	The identity matrix with the dimension of clusters
$I_n \in \mathbb{R}^{n \times n}$	The identity matrix with the dimension of samples
$\text{Tr}(\cdot)$	The trace of matrix
$\ \cdot\ _{\otimes}$	The tensor nuclear norm

based on self-representation learning construct the affinity matrices by forcibly enforcing the non-negative constraints, which destroys the inherent relationships between data. The affine constraints used by these methods are not flexible enough. Scaled simplex representation offers a promising solution to these problems. The mathematical model is defined as follows:

$$\begin{aligned} \min_{C^v} \sum_{v=1}^{n_v} \|X^v - X^v C^v\|_F^2 + \alpha \|C^v\|_F^2 \\ \text{s.t. } C^v \geq 0, \mathbf{1}^T C^v = s \mathbf{1}^T \end{aligned} \quad (1)$$

where $X^v \in \mathbb{R}^{d^v \times n}$ is the original data matrix, and $C^v \in \mathbb{R}^{n \times n}$ is the affinity matrix.

3. Proposed method

3.1. Notations

We first present the main notations used in this article. For a matrix A , the Frobenius norm and nuclear norm are represented as follows: $\|A\|_F = \sqrt{\sum_{ij} |A_{ij}|^2}$ and $\|A\|_* = \sum_i \sigma_i(A)$, where $\sigma_i(A)$ is the i -th singular value of A . Additionally, the $l_{2,1}$ -norm of matrix A is defined as $\|A\|_{2,1} = \sum_i \sqrt{\sum_j |A_{ij}|^2}$. The transpose and inverse of matrix A are represented by A^T and A^{-1} , respectively. For a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, it can be constructed through the stacking operation $\Phi(\cdot)$, expressed as $\mathcal{A} = \Phi(A^1, A^2, \dots, A^{n_3})$, where each A^v corresponds to the v -th frontal slice of $\mathcal{A}$. The discrete Fourier transform is represented by $\bar{\mathcal{A}} = \text{fft}(\mathcal{A}, [], 3)$, with its inverse transformation given by $\mathcal{A} = \text{ifft}(\bar{\mathcal{A}}, [], 3)$. The main notations throughout this article are reported in Table 1.

The following will present the definitions of the tensor singular value decomposition (t-SVD) and t-SVD based tensor nuclear norm (t-TNN).

Definition 1 (t-SVD (Kilmer & Martin, 2011)) Given a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its t-SVD is defined as follows:

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \quad (2)$$

where $\mathcal{S}$ is an f-diagonal tensor with each frontal slice being diagonal, and $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal tensors.

According to Kilmer et al. (2013), the t-SVD problem can be efficiently solved using matrix SVD in the Fourier domain. Specifically, the decomposition can be expressed as $\bar{\mathcal{A}}^v = \bar{\mathcal{U}}^v \bar{\mathcal{S}}^v \bar{\mathcal{V}}^{vT}$, where $v = 1, \dots, n_3$.

Definition 2 (t-TNN (Zhang et al., 2014)) For a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the t-TNN is formulated as follows:

$$\|\mathcal{A}\|_{\otimes} = \sum_{v=1}^{n_3} \|\bar{\mathcal{A}}^v\|_{\otimes} = \sum_{v=1}^{n_3} \sum_{i=1}^{\min(n_1, n_2)} \sigma_i(\bar{\mathcal{A}}^v) \quad (3)$$

where $\sigma_i(\bar{A}^v)$ denotes the i -th largest singular value of $\bar{A}^v$.

In addition, it is possible to define the weighted t-TNN as follows [Gao et al. \(2020\)](#):

$$\|\mathcal{A}\|_{\omega, \otimes} = \sum_{v=1}^{n_3} \sum_{i=1}^{\min(n_1, n_2)} \omega_i \sigma_i(\bar{A}^v) \quad (4)$$

where ω_i is the weighting factor of $\sigma_i(\bar{A}^v)$.

3.2. Clean affinity matrix based MUFS

In feature selection tasks, a common approach is to utilize the self-representation property of data to construct high-quality affinity matrices, in which each data point is represented by other data points in the same subspace via linear reconstruction. The formula is as follows:

$$\min_{W^v, C^v} \sum_{v=1}^{n_v} \|W^{vT} X^v - W^{vT} X^v C^v\|_F^2 + \gamma \text{rank}(C^v) \quad (5)$$

$$\text{s.t. } W^{vT} X^v X^{vT} W^v = I_c$$

where $W^v \in \mathbb{R}^{d^v \times c}$ is the projection matrix, $C^v \in \mathbb{R}^{n \times n}$ is the affinity matrix, γ is a hyperparameter, and $\text{rank}(C^v)$ is used to avoid trivial solutions. By applying the constraint $W^{vT} X^v X^{vT} W^v = I_c$, redundant features can be avoided in the selection process. However, most self-representation learning methods do not take into account the effect of noise in the original data. Inspired by robust PCA ([Candès et al., 2011](#)), we decompose the affinity matrix into a clean component and a noise component, which is formulated as follows:

$$C^v = Z^v + E^v \quad (6)$$

where $Z^v \in \mathbb{R}^{n \times n}$ is the clean affinity matrix, and $E^v \in \mathbb{R}^{n \times n}$ is the noise matrix.

Then, we impose the $\ell_{2,1}$ -norm on the W^v to encourage row-wise sparsity, enabling the selection of discriminative features, and on the E^v to separate and suppress sample-level noise components, thereby isolating the clean data structure from corrupted observations. [Eq. \(5\)](#) is further formulated as

$$\begin{aligned} \min_{W^v, Z^v, E^v} \sum_{v=1}^{n_v} & \|W^{vT} X^v - W^{vT} X^v (Z^v + E^v)\|_F^2 + \alpha \|W^v\|_{2,1} \\ & + \beta \|E^v\|_{2,1} + \gamma \text{rank}(Z^v) \\ \text{s.t. } & W^{vT} X^v X^{vT} W^v = I_c \end{aligned} \quad (7)$$

where α and β are trade-off parameters. Specifically, α controls the sparsity of W^v for feature selection, while β controls the sparsity of E^v to suppress noise and enhance robustness.

3.3. Scaled simplex constraint

Enforcing forced non-negativity constraints on the affinity matrices may distort the inherent relationships among data points. To address these issues and enhance the flexibility of the clean affinity matrices, we adopt the scaled simplex representation ([Xu et al., 2019](#)), which is formulated as follows:

$$\begin{aligned} \min_{W^v, Z^v, E^v} \sum_{v=1}^{n_v} & \|W^{vT} X^v - W^{vT} X^v (Z^v + E^v)\|_F^2 + \alpha \|W^v\|_{2,1} \\ & + \beta \|E^v\|_{2,1} + \gamma \text{rank}(Z^v) \\ \text{s.t. } & W^{vT} X^v X^{vT} W^v = I_c, Z^v \geq 0, \mathbf{1}^T Z^v = s \mathbf{1}^T \end{aligned} \quad (8)$$

where the constraint $Z^v \geq 0$ enhances the discriminative ability of Z^v , and s is a preset scaling parameter that determines the sum of each coefficient vector z_i^v , thereby balancing the sparsity and connectivity of the affinity matrix. Specifically, the scaled simplex constraint enforces the entries of z_i^v to be non-negative and their sum equal to s . Smaller values of s lead to sparser and more discriminative connections, whereas larger values of s yield denser and more connected affinity structures.

By tuning this preset value, the structure of Z^v can flexibly capture data relationships. Moreover, the scaled simplex constraint smoothly transforms negative affinities into non-negative ones, avoiding the abrupt distortions caused by hard non-negativity enforcement. Therefore, this constraint plays a crucial role in improving the flexibility and discriminative capability of MUFS.

3.4. Rank constraint

Similar to the clustering task, the data is divided into c clusters. The optimal neighbor assignment ensures that the connected components in the data accurately correspond to these clusters. To achieve this, we impose a rank constraint ([Nie et al., 2014](#)) to ensure that the learned clean affinity matrix exhibits an optimal structure. This can be formulated as follows:

$$\text{rank}(L_{Z^v}) = n - c \quad (9)$$

where the Laplacian matrix L_{Z^v} is defined as $L_{Z^v} = D^v - \frac{Z^v + Z^{vT}}{2}$, in which D^v is a diagonal degree matrix with the diagonal entries $d_{ii}^v = \frac{1}{2} \sum_j (z_{ij}^v + z_{ji}^v)$. Based on the theory of spectral graph ([Chung, 1997](#); [Fan, 1949](#); [Mohar et al., 1991](#); [Nie et al., 2016](#)), the following theorem can be obtained.

Theorem 1. *The multiplicity of the eigenvalue 0 of the Laplacian matrix L_{Z^v} of the graph generated by the matrix Z^v equals the number of connected components in the graph.*

Consequently, [Eq. \(9\)](#) guarantees that there are c connected components in Z^v , ensuring that the affinity structure aligns with the desired cluster partition. In the context of feature selection, this constraint enforces a block-diagonal structure in the sample affinity matrix, which helps group samples according to their intrinsic relationships and preserves the global manifold structure of the data. It effectively prevents unrelated samples from being connected, thereby improving the discriminability of the selected features. Due to the difficulty of directly solving the rank constraint, according to Ky Fan theorem ([Fan, 1949](#)), the rank constraint can be transformed into the following equivalence problem:

$$\min_{F^v} \text{Tr} \left(F^{vT} L_{Z^v} F^v \right) \text{ s.t. } F^{vT} F^v = I_c \quad (10)$$

where $F^v \in \mathbb{R}^{n \times c}$ is the indicator matrix corresponding to the Laplacian matrix L_{Z^v} . Then, the following objective is obtained.

$$\begin{aligned} \min_{W^v, Z^v, E^v, F^v} \sum_{v=1}^{n_v} & \|W^{vT} X^v - W^{vT} X^v (Z^v + E^v)\|_F^2 + \alpha \|W^v\|_{2,1} \\ & + \beta \|E^v\|_{2,1} + 2\gamma \text{Tr} \left(F^{vT} L_{Z^v} F^v \right) \\ \text{s.t. } & W^{vT} X^v X^{vT} W^v = I_c, Z^v \geq 0, \mathbf{1}^T Z^v = s \mathbf{1}^T, \\ & F^{vT} F^v = I_c \end{aligned} \quad (11)$$

where γ balances the rank constraint to maintain the intrinsic structure of data.

3.5. Weighted low-rank tensor constraint

To capture higher-order relationships among views, A practical method is to stack the individual view matrix Z^v into a tensor $\mathcal{Z}$. However, in real-world applications, the contributions of different views are usually unequal. As [Gao et al. \(2020\)](#), we assign different weights for each view to better capture the relationships between views. It can be formulated as

$$\|\mathcal{Z}\|_{\omega, \otimes} = \sum_{v=1}^{n_3} \sum_{i=1}^{\min(n_1, n_2)} \omega_i \sigma_i(\bar{\mathcal{Z}}^v) \quad (12)$$

3.6. Overall objective function

In our framework, these components work in a complementary manner. Specifically, it first separates noise from the original data to obtain a clean representation, thereby alleviating the influence of noisy features. Then, a scaled simplex constraint is imposed to flexibly capture the global relationships among data points based on this clean representation. Furthermore, a rank constraint is applied to ensure that the learned affinity matrix precisely corresponds to the c clusters in the data. Finally, a weighted tensor is constructed from the high-quality affinity matrices to explore high-order relationships across views. These components complement each other to progressively improve the quality of the affinity matrix and further exploit the high-quality affinity matrix to construct a tensor that captures high-order relationships across views, thereby enhancing the performance of multi-view feature selection. Thus, by combining Eqs. (11) and (12), the overall objective function of our RSTR method is formulated as follows:

$$\min_{\phi} \sum_{v=1}^{n_v} \|W^{vT} X^v - W^{vT} X^v(Z^v + E^v)\|_F^2 + \alpha \|W^v\|_{2,1} + \beta \|E^v\|_{2,1} + 2\gamma \text{Tr}(F^{vT} L_{Z^v} F^v) + \|\mathcal{J}\|_{\omega, \otimes} \quad (13)$$

$$\text{s.t. } W^{vT} X^v X^{vT} W^v = I_c, \quad Z^v \geq 0, \quad \mathbf{1}^T Z^v = s \mathbf{1}^T,$$

$$F^{vT} F^v = I_c$$

where $\phi = \{W^v, Z^v, E^v, F^v, \mathcal{J}\}_{v=1}^{n_v}$.

3.7. Optimization

It is notable that simultaneously solving the multiple unknown variables in Eq. (13) is challenging. Therefore, the overall optimization problem is decomposed into several subproblems, which are solved using the augmented Lagrangian multiplier method combined with alternating direction minimization (ALM-ADM) (Bertsekas, 2014).

First, an auxiliary tensor $\mathcal{J}$ is introduced into Eq. (13), Eq. (13) is rewritten as

$$\min_{\phi} \sum_{v=1}^{n_v} \|W^{vT} X^v - W^{vT} X^v(Z^v + E^v)\|_F^2 + \alpha \|W^v\|_{2,1} + \beta \|E^v\|_{2,1} + 2\gamma \text{Tr}(F^{vT} L_{Z^v} F^v) + \|\mathcal{J}\|_{\omega, \otimes} \quad (14)$$

$$\text{s.t. } W^{vT} X^v X^{vT} W^v = I_c, \quad Z^v \geq 0, \quad \mathbf{1}^T Z^v = s \mathbf{1}^T,$$

$$F^{vT} F^v = I_c, \quad \mathcal{Z} = \mathcal{J}$$

where $\phi = \{W^v, Z^v, E^v, F^v, \mathcal{J}\}_{v=1}^{n_v}$. Then, an augmented Lagrangian function is as follows:

$$\min_{\phi} \sum_{v=1}^{n_v} \|W^{vT} X^v - W^{vT} X^v(Z^v + E^v)\|_F^2 + \alpha \|W^v\|_{2,1} + \beta \|E^v\|_{2,1} + 2\gamma \text{Tr}(F^{vT} L_{Z^v} F^v) + \|\mathcal{J}\|_{\omega, \otimes} + \frac{\mu}{2} \|\mathcal{J} - (\mathcal{Z} + \frac{\mathcal{Q}}{\mu})\|_F^2 \quad (15)$$

$$\text{s.t. } W^{vT} X^v X^{vT} W^v = I_c, \quad Z^v \geq 0, \quad \mathbf{1}^T Z^v = s \mathbf{1}^T,$$

$$F^{vT} F^v = I_c$$

where $\mu > 0$ serves as a penalty parameter, and $\mathcal{Q}$ denotes the Lagrangian multiplier.

Then, Eq. (15) is decomposed into the following subproblems, where each variable is updated while keeping the others fixed. The steps are as follows:

Update W^v : Fix Z^v , E^v , F^v , and $\mathcal{J}$, Eq. (15) is reduced to the following equation:

$$\min_{W^v} \sum_{v=1}^{n_v} \|W^{vT} X^v - W^{vT} X^v(Z^v + E^v)\|_F^2 + \alpha \|W^v\|_{2,1} \quad (16)$$

$$\text{s.t. } W^{vT} X^v X^{vT} W^v = I_c$$

Algorithm 1 The proposed RSTR method.

Input: The original data matrices $\{X^v\}_{v=1}^{n_v}$, the hyperparameters α , β , and γ , the scaled value s , the proportion of selected features k , and the number of clusters c .

Initialize: Construct feature selection matrices $\{W^v\}_{v=1}^{n_v}$, clean affinity matrices $\{Z^v\}_{v=1}^{n_v}$, noise matrices $\{E^v\}_{v=1}^{n_v}$, and indicator matrices $\{F^v\}_{v=1}^{n_v}$. Initialize tensor $\mathcal{Q} = \mathcal{J} = 0$.

Update:

repeat

Update W^v via Eq. (20);

Update Z^v via Eq. (23);

Update E^v via Eq. (28);

Update F^v via Eq. (29);

Update $\mathcal{J}$ via Eq. (30);

Update $\mathcal{Q}$ via Eq. (40);

Update μ via Eq. (40);

until convergence

Output: The learned feature selection matrix W .

Feature selection: Sort all features of the multi-view data in descending order according to $\|W_i\|_2$, and the top- k ranked features are selected accordingly.

Table 2

Statistics of multi-view datasets in the experiments.

Datasets	Samples	Views	Clusters	Dimensions
BBCsport	544	2	5	3183, 3203
Coil20	1440	3	20	944, 324, 512
HW2sources	2000	2	10	784, 256
MSRC-v1	210	5	7	1302, 48, 512, 256, 210
ORL	400	3	40	4096, 3304, 6750
UCI	2000	3	10	240, 76, 6
Outscene	2688	4	8	512, 432, 256, 48
WebKB	203	3	4	1703, 230, 230
Animal	10,158	2	50	4096, 4096

To optimize Eq. (16), an auxiliary diagonal matrix P^v is introduced, with its diagonal elements given by

$$P_{ii}^v = \frac{1}{\max(2\|W_i^v\|_2, \epsilon)} \quad (17)$$

Thus, W^v is optimized as follows:

$$(X^v U^v X^{vT} + \alpha P^v) W^v = \Lambda^v X^v X^{vT} W^v \quad (18)$$

where Λ^v is a diagonal matrix with diagonal elements as eigenvalues, and $U^v = (I_n - (Z^v + E^v))(I_n - (Z^v + E^v))^T$.

However, since the number of features is often much larger than the number of samples, the conditions required to solve Eq. (18) are difficult to satisfy in the real world. To this end, as expressed in Du and Shen (2015), we reformulate the optimization problem for the variable W^v as follows:

$$\min_{W^v} \|Y^v - X^{vT} W^v\|_F^2 + \alpha \|W^v\|_{2,1} \quad (19)$$

where Y^v consists of the eigenvectors of U^v corresponding to the smallest c eigenvalues. Referring to Liang et al. (2017), the following formula is obtained:

$$W^v = (X^v X^{vT} + \alpha P^v)^{-1} X^v Y^v \quad (20)$$

Update Z^v : Fix W^v , E^v , F^v , and $\mathcal{J}$, Eq. (15) is reduced to

$$\min_{Z^v} \sum_{v=1}^{n_v} \|W^{vT} X^v - W^{vT} X^v(Z^v + E^v)\|_F^2 + 2\gamma \text{Tr}(F^{vT} L_{Z^v} F^v) + \frac{\mu}{2} \|J^v - (Z^v + \frac{\mathcal{Q}^v}{\mu})\|_F^2 \quad (21)$$

$$\text{s.t. } Z^v \geq 0, \mathbf{1}^T Z^v = s \mathbf{1}^T$$

Table 3

ACC (mean±std (%)) of compared methods on eight real-world datasets.

Dataset	Baseline	LS	TLR-MFS	JMVFG	CE-UMFS	PTFS	GCDUFS	WLTL	RSTR
BBCsport	48.03±10.98	43.50±6.65	60.78±9.79	47.66±6.10	62.10±6.78	60.69±7.50	55.95±13.69	62.53±9.94	65.77 ± 9.21
Coil20	62.03±3.51	57.35±2.40	59.43±2.80	69.23±3.96	70.01±2.79	59.54±5.40	58.79±3.87	62.61±3.20	76.48 ± 2.35
HW2sources	62.70±4.05	53.55±0.99	46.08±0.89	79.57±3.94	68.02±0.89	78.07±9.36	46.55±2.55	75.89±8.58	87.99 ± 0.33
MSRC-v1	34.43±2.35	31.86±0.71	65.95±6.58	84.73±4.58	80.55±2.00	80.25±9.85	78.79±7.75	75.52±16.13	89.00 ± 5.19
ORL	54.46±3.06	50.46±2.59	54.09±2.26	59.66±2.40	60.30±2.05	61.97±3.46	51.76±3.49	62.60±1.84	63.65 ± 2.06
UCI	39.24±0.11	39.22±0.10	39.30±0.15	73.93±4.15	85.81±0.40	83.24±6.16	56.87±3.28	81.71±3.06	95.10 ± 0.08
Outscene	33.12±0.11	33.39±0.14	48.11±0.28	65.09±2.84	56.83±0.01	61.73±4.98	56.17±1.19	62.54±3.67	68.52 ± 0.46
WebKB	67.44±3.49	66.26±2.43	67.86±2.50	72.31±9.24	75.10±1.27	70.80±7.82	71.06±6.59	59.51±12.77	77.88 ± 2.10
Animal	64.42±1.74	64.44±1.33	62.26±1.42	67.63±1.77	67.20±1.75	67.08±1.95	N/A	64.88±1.57	68.17 ± 1.85

Table 4

NMI (mean±std (%)) of compared methods on eight real-world datasets.

Dataset	Baseline	LS	TLR-MFS	JMVFG	CE-UMFS	PTFS	GCDUFS	WLTL	RSTR
BBCsport	21.15±15.26	18.34±5.30	40.78±12.70	16.60±7.05	41.47±9.12	13.63±9.35	37.90±19.55	44.85±15.84	46.61 ± 12.81
Coil20	73.39±2.00	69.35±1.17	70.85±1.00	79.71±1.73	79.93±1.40	72.51±2.55	70.74±2.03	75.73±1.93	83.85 ± 1.55
HW2sources	57.39±3.81	46.52±1.11	41.63±1.46	68.09±2.60	57.54±0.58	76.09±5.36	41.37±1.62	73.46±6.63	77.82 ± 0.45
MSRC-v1	20.62±2.30	18.42±1.04	58.64±5.07	75.41±4.45	74.53±1.53	75.11±5.90	71.27±4.70	67.98±12.35	80.53 ± 4.06
ORL	73.66±1.42	71.78±1.68	73.34±1.42	77.31±1.13	76.47±1.17	79.61±1.85	71.51±1.86	80.57±1.19	79.37±1.50
UCI	46.74±0.01	46.73±0.01	46.71±0.04	66.14±2.12	73.72±0.46	77.59±2.65	50.41±2.45	79.59±1.17	87.92 ± 0.17
Outscene	17.65±0.04	17.78±0.13	35.14±0.35	51.17±1.08	41.95±0.01	50.76±1.68	31.70±1.29	49.45±2.95	51.73 ± 0.21
WebKB	38.55±2.94	39.93±2.34	40.50±2.07	45.19±6.74	33.83±3.45	33.76±5.46	40.21±4.36	20.09±11.92	37.33±4.73
Animal	73.53±0.52	74.01±0.43	71.68±0.42	75.98±0.37	75.94±0.50	75.08±0.32	N/A	74.03±0.39	76.25 ± 0.57

The derivative of Eq. (21) concerning Z^v is obtained as

$$\hat{Z}^v = (2X^{vT} W^v W^{vT} X^v + \mu I_n)^{-1} (2X^{vT} W^v W^{vT} X^v (I_n - E^v) + \mu(J^v - \frac{Q^v}{\mu}) - \gamma M^v) \quad (22)$$

where $M_{ij}^v = \|F_i^v - F_j^v\|_2^2$. After updating $\hat{Z}^v$, we obtain

$$Z^v = \min_{Z^v \geq 0, 1^T Z^v = 1^T} \|Z^v - \hat{Z}^v\|_F^2 \quad (23)$$

By considering the constraints in Eq. (23), the solution can be formulated as follows:

$$Z_{ij}^v = \begin{cases} (\tilde{Z}_{ij}^v + \delta)_+ & i \neq j \\ 0, & i = j \end{cases} \quad (24)$$

where the function $(f)_+$ ensures data non-negative, and δ is updated as

$$\delta = \frac{1}{n-1} - \frac{1}{n-1} \sum_{j=1, j \neq i}^n \tilde{Z}_{ij}^v. \quad (25)$$

Update E^v : Fix W^v , Z^v , F^v and J , Eq. (15) is reduced to

$$\min_{E^v} \sum_{v=1}^{n_v} \|W^{vT} X^v - W^{vT} X^v (Z^v + E^v)\|_F^2 + \beta \|E^v\|_{2,1} \quad (26)$$

The derivative of Eq. (26) concerning E^v is obtained as

$$E^v = (2X^{vT} W^v W^{vT} X^v + 2\beta O^v)^{-1} (2X^{vT} W^v W^{vT} X^v (I_n - Z^v)) \quad (27)$$

Similar to Eq. (17), O^v is a diagonal matrix with its diagonal elements given by

$$O_{ii}^v = \frac{1}{\max(2\|E_{i:}^v\|_2, \epsilon)} \quad (28)$$

Update F^v : Fix W^v , Z^v , E^v , and J , Eq. (15) is reduced to

$$\text{Tr}(F^{vT} L_{Z^v} F^v) \quad \text{s.t.} \quad F^{vT} F^v = I_c \quad (29)$$

The solution is given by the c eigenvectors associated with the c smallest eigenvalues of L_{Z^v} .

Update J : Fix W^v , Z^v , E^v , and F^v , Eq. (15) is reduced to

$$\min_J \|J\|_{\omega, \otimes} + \frac{\mu}{2} \|J - (Z + \frac{Q}{\mu})\|_F^2 \quad (30)$$

To derive the solution of Eq. (30), two theorems are introduced as follows:

Theorem 2. For a matrix $T \in \mathbb{R}^{m \times n}$ with singular value decomposition $T = U * S * V^T$, let $\theta > 0$, and $r = \min(m, n)$, the standard weighted nuclear norm minimization problem is defined as

$$\min_H \frac{1}{2} \|H - T\|_F^2 + \theta \|H\|_{\omega, *}, \quad (31)$$

Accordingly, the optimal solution to Eq. (31) can be expressed as

$$H^* = \Gamma_{\theta * \omega}[T] = U P_{\theta * \omega}(T) V^T \quad (32)$$

where $P_{\theta * \omega}(T) = \text{diag}(\delta_1, \delta_2, \dots, \delta_r)$, and $\delta_i = \text{sign}(\sigma_i(T)) \max(\sigma_i(T) - \theta, 0)$.

Theorem 3. Given a tensor $F \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ with $r = \min(n_1, n_2)$ and its t -SVD $F = U * T * V^T$, we obtain the following model according to Theorem 2:

$$\min_H \frac{1}{2} \|H - F\|_F^2 + \theta \|H\|_{\omega, \otimes} \quad (33)$$

Then, the optimal solution to Eq. (33) is given by

$$H^* = \Gamma_{\theta * \omega}(F) = U * \text{ifft}(P_{\theta * \omega}(\bar{F})) * V^T \quad (34)$$

where $\bar{F} = \text{fft}(F, [1, 3])$ denotes the tensor transformed into the Fourier domain along the third mode, and $P_{\theta * \omega}(\bar{F}^{(k)})$ represents the k -th frontal slice of $P_{\theta * \omega}(\bar{F})$.

Proof. Following Liu and Song (2024), Eq. (33) can be reformulated in the Fourier domain as follows:

$$\bar{H}^* = \min_{\bar{H}} \frac{1}{2} \|\bar{H} - \bar{F}\|_F^2 + \sum_{k=1}^{n_3} \sum_{j=1}^r \theta * \omega_j * \sigma_j(\bar{H}^{(k)}) \quad (35)$$

To solve Eq. (35), it can be reformulated as follows:

$$\min_{\bar{H}} \sum_{k=1}^{n_3} \left(\frac{1}{2} \|\bar{H}^{(k)} - \bar{F}^{(k)}\|_F^2 + \sum_{j=1}^r \theta * \omega_j * \sigma_j(\bar{H}^{(k)}) \right) \quad (36)$$

Then, Eq. (36) can be decomposed into n_3 independent subproblems as follows:

$$\bar{H}^{(k)*} = \min_{\bar{H}^{(k)}} \frac{1}{2} \|\bar{H}^{(k)} - \bar{F}^{(k)}\|_F^2 + \sum_{j=1}^r \theta * \omega_j * \sigma_j(\bar{H}^{(k)}) \quad (37)$$

Based on Theorem 1, the solution to problem (37) can be written as $\bar{H}^{(k)*} = \Gamma_{\theta * \omega}[\bar{F}^{(k)}] = \bar{U}^{(k)} P_{\theta * \omega}(\bar{F}^{(k)}) \bar{V}^{(k)T}$, where $\bar{H}^{(k)*}$, $\bar{U}^{(k)}$, and $\bar{V}^{(k)}$

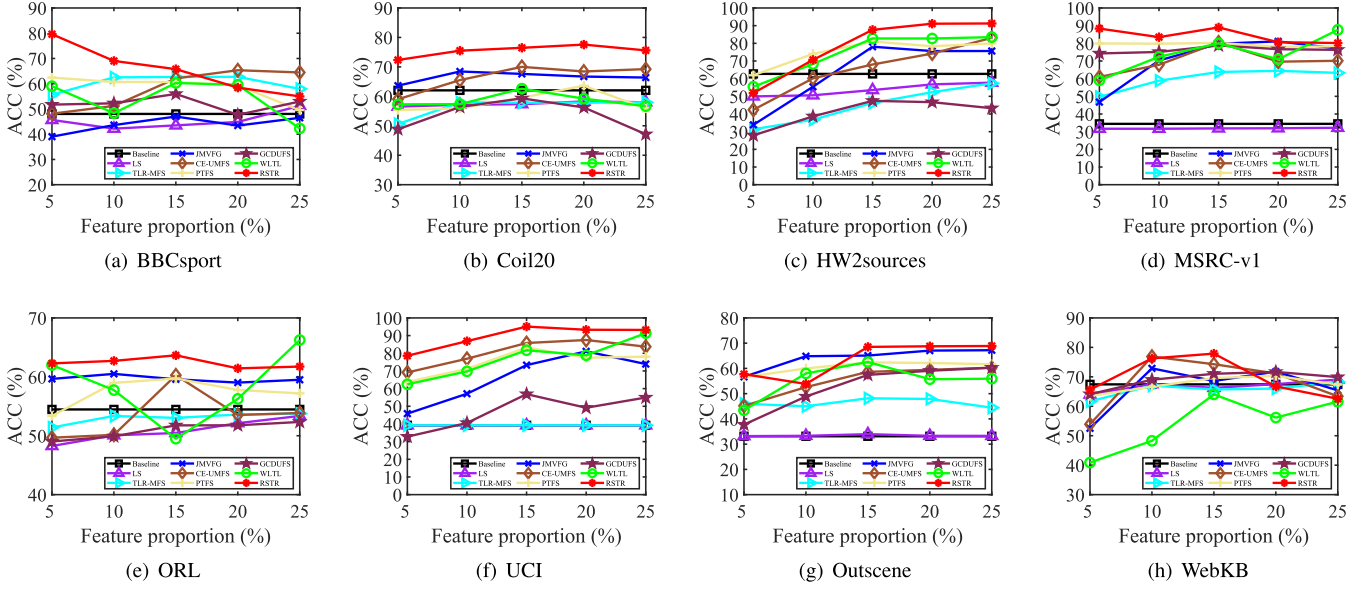


Fig. 2. ACC performance on eight datasets under different selection proportions.

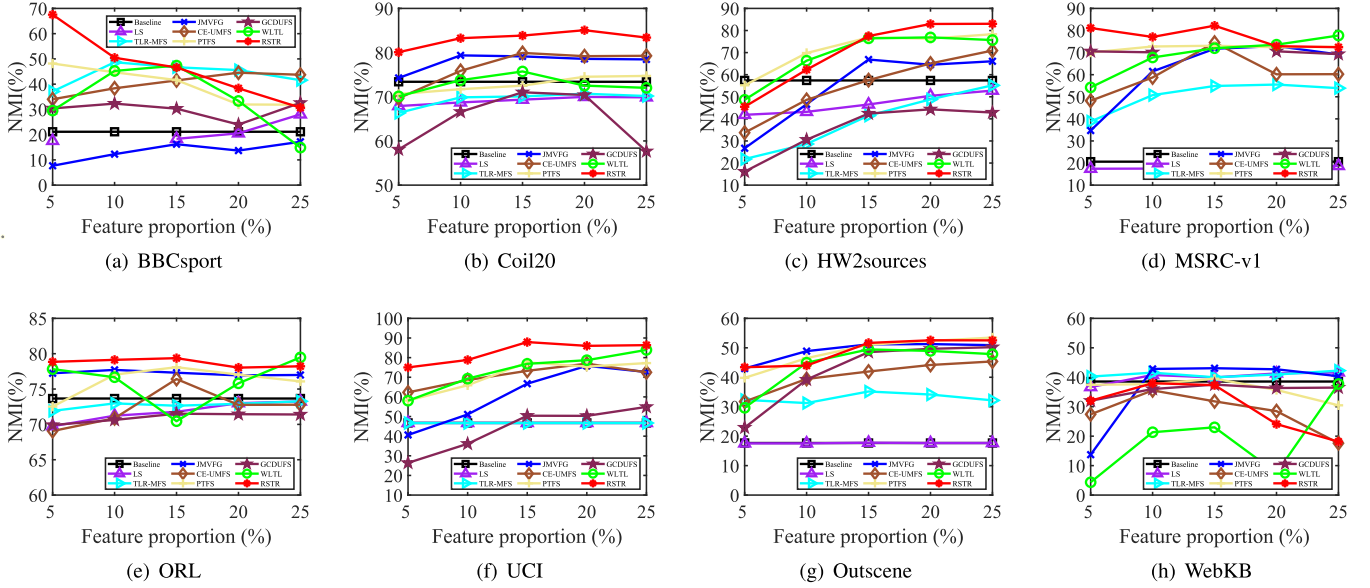


Fig. 3. NMI performance on eight datasets under different selection proportions.

Table 5
Computational time (in seconds) of different methods on nine datasets.

Datasets	LS	TLR-MFS	JMVFG	CE-UMFS	PTFS	GCDUFS	WLTL	RSTR
BBCsport	0.07	2.55	5.63	13.73	23.43	79.76	11.38	9.05
Coil20	0.04	1.04	5.85	4.28	41.76	115.22	3.22	41.71
HW2sources	0.03	1.20	7.25	4.13	56.56	191.87	2.21	64.08
MSRC-v1	0.01	0.41	0.96	0.95	2.37	5.48	3.85	0.96
ORL	0.10	22.24	48.36	36.62	150.20	502.17	154.13	25.39
UCI	0.01	3.48	6.18	2.23	78.54	304.41	2.00	295.57
Outscene	0.06	6.90	12.89	8.31	232.21	953.34	6.59	93.06
WebKB	0.01	0.45	0.64	1.24	3.61	6.57	3.23	1.18
Animal	1.14	107.73	406.45	345.97	2572.75	N/A	1200.74	4367.23

denote the k -th frontal slices of $\bar{H}^*$, $\bar{U}$, and $\bar{V}$, respectively. Thus, we obtain

$$\mathcal{H}^* = \Gamma_{\theta^* \omega}[F] = \mathcal{U} * \text{ifft}(P_{\theta^* \omega}(\bar{F})) * \mathcal{V}^T \quad (38)$$

where $\mathcal{U} = \text{ifft}(\bar{U}, [\cdot], 3)$ and $\mathcal{V} = \text{ifft}(\bar{V}, [\cdot], 3)$.

Thus, based on Theorem 2, the solution of Eq. (30) is expressed as follows:

$$\mathcal{J}^* = \Gamma_{\mu}^{\ell_{\theta^* \omega}}[\mathcal{Z} + \frac{\mathcal{Q}}{\mu}] \quad (39)$$

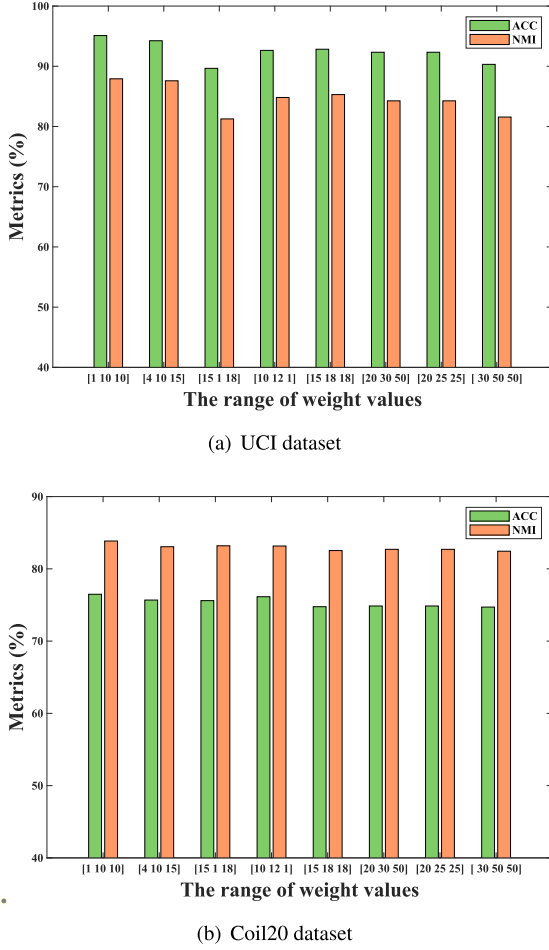


Fig. 4. Clustering performance of RSTR with different weights.

Update μ and Q : We update the penalty parameter μ and the Lagrangian multiplier Q using the following update formulas:

$$\begin{aligned}\mu &= \rho * \mu \\ Q &= Q + \mu(Z - J)\end{aligned}\quad (40)$$

By adjusting the values of ρ , it can be ensured that the algorithm converges smoothly. Algorithm 1 provides the detailed procedures of RSTR.

3.8. Complexity analysis

The time complexity of RSTR is primarily determined by solving the following subproblems: W^v , Z^v , E^v , F^v , and J . The update of W^v incurs a cost of $\mathcal{O}(cn^2 + d^v)^3$. The computation of Z^v and E^v demand $\mathcal{O}(n_v n^3)$. The computation for updating F^v is $\mathcal{O}(n_v c n^2)$. The computation for updating J is $\mathcal{O}(n_v n^2 \log n)$. Therefore, the overall complexity of RSTR is nearly $\mathcal{O}(n_v n^3 + d^v)^3$.

4. Experiments

In this section, we conduct a series of experiments on eight real-world datasets and compare RSTR with eight other methods to validate its effectiveness.

4.1. Datasets

In the experiments, we use the following eight widely used real-world multi-view datasets. The detailed information is provided in Table 2.

- **BBCsport** Xu et al. (2025) consists of news articles from the BBCsport website, covering various sports categories.
- **Coil20** Duan et al. (2025) comprises 20 distinct categories, each captured from 72 different viewpoints at 5-degree intervals under controlled lighting and a uniform background.
- **HW2sources** Wang et al. (2025a) integrates two handwritten digital image sources (i.e., USPS and MNIST), offering two different styles of image representation for each sample.
- **MSRC-v1** Zhou and Song (2024) consists of a variety of images of real-world objects, and each sample contains five different types of feature representations covering a range of information such as color, texture, and shape.
- **ORL** Cai et al. (2024) collects facial images from 40 individuals, and each contains multiple images with slight variations in expression and posture.
- **UCI** Chen et al. (2022) consists of handwritten digit images, and each sample is represented by three different types of extracted features.
- **Outscene** Cao and Xie (2024a) contains eight categories of outdoor scene images. Each image is described using different visual features, including color, texture, and edge information.
- **WebKB** Wang and Song (2025) consists of webpages from four universities, with each page represented by three views: content features, inbound links, and outbound links.
- **Animal** Zhou and Song (2024) consists of 10,158 images from 50 categories. Each sample has two views, both represented by 4096-dimensional features.

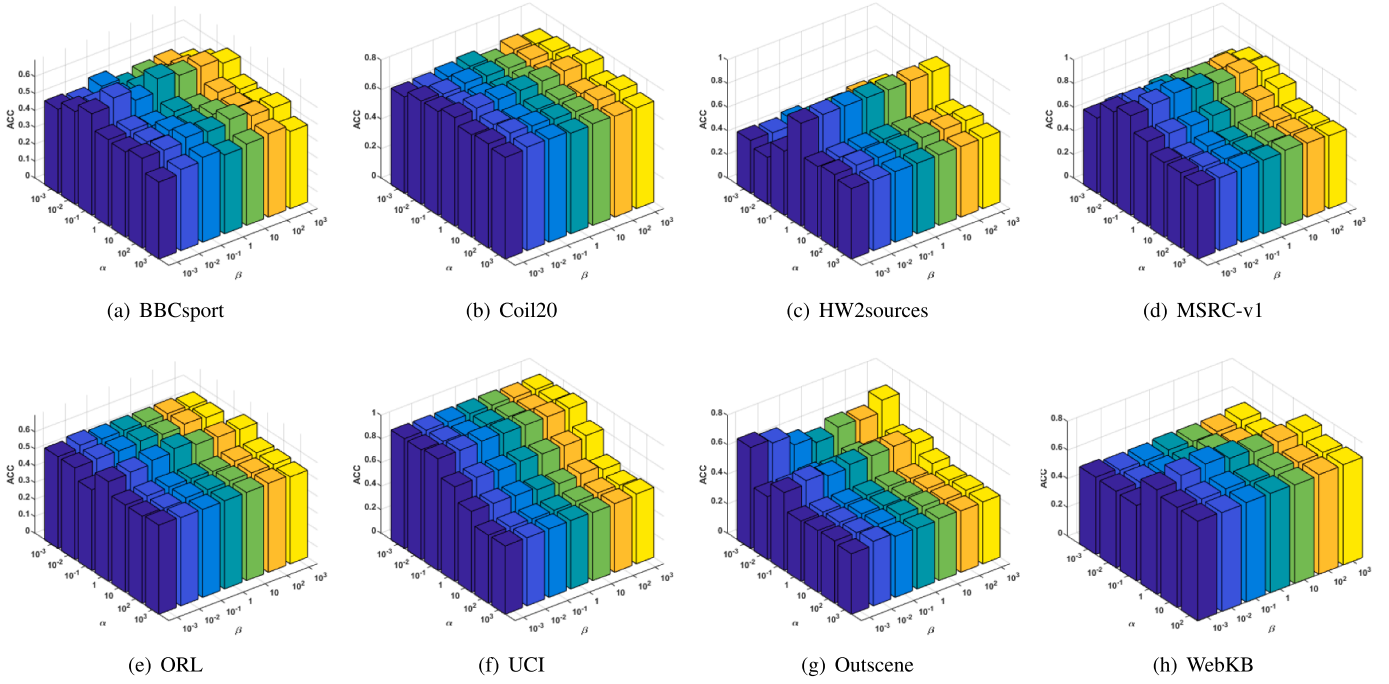
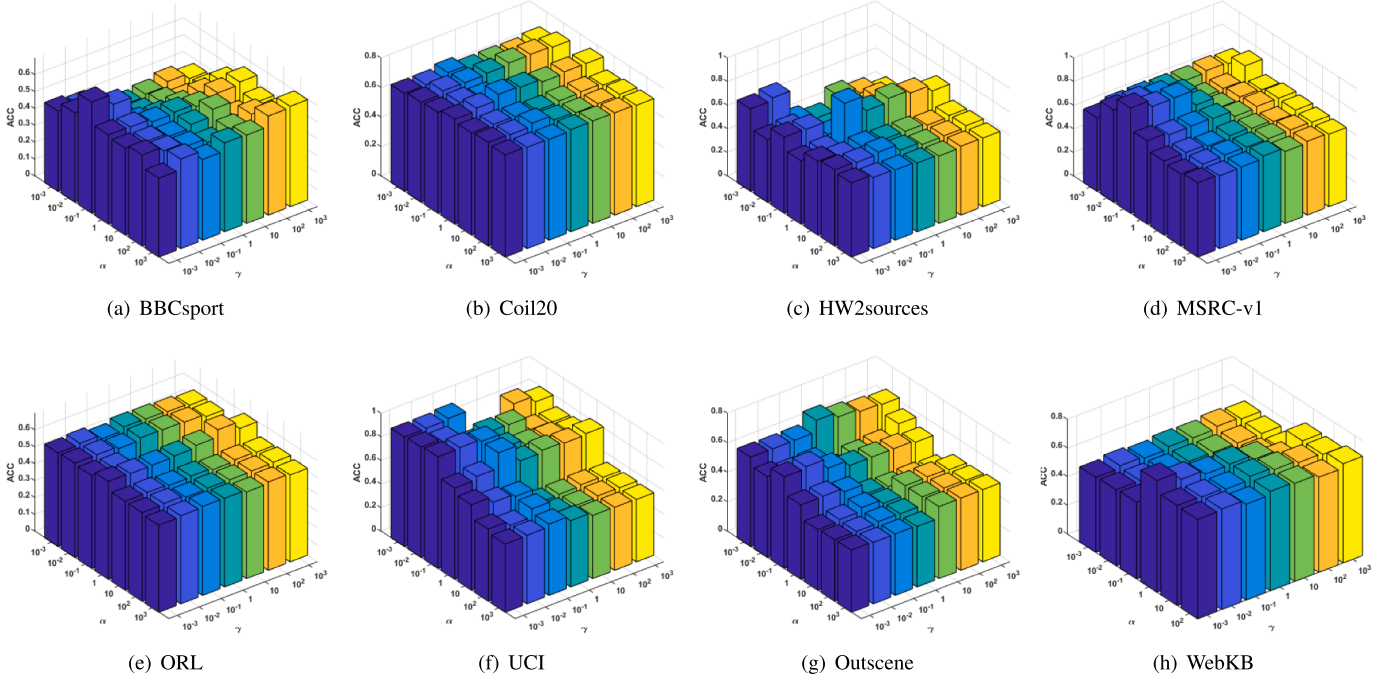
4.2. Compared methods

To assess the effectiveness of RSTR, we compare it with eight different UFS methods. The detailed description of these methods is provided below.

- **Baseline** is a baseline method that selects all the original features.
- **LS** He et al. (2005) is a classic single-view UFS algorithm. In our experiments, we perform feature selection by combining features from all views into a unified view.
- **TLR-MFS** Yuan et al. (2022) is a low-rank tensor constrained graph embedding model that builds separate graph structures for each view and captures consistency between views.
- **JMVFG** Fang et al. (2023a) combines orthogonal decomposition with graph structure, and introduces a local structure preserving term to construct a feature selection model.
- **CE-UMFS** Zhou and Song (2024) is a feature selection model based on matrix decomposition. It aims to simultaneously mine consistency and exclusivity, while maintaining exclusivity between views using the Hilbert-Schmidt independence criterion.
- **PTFS** Cao and Xie (2024a) optimizes a low-rank tensor to mine higher-order relationships. It also leverages discriminative partitioning information, and introduces a self-paced strategy for feature selection.
- **GCDUFS** Xu et al. (2025) constrains the consensus matrix using graph regularization and employs a kernel function to measure the dependence between the data in Hilbert space.
- **WLTL** Wang et al. (2025b) explores multi-view consistency by stacking clustering indicator matrices into a tensor and applying a weighted tensor to capture higher-order relationships among different views. It also adaptively learns view weights to enhance the clustering performance.

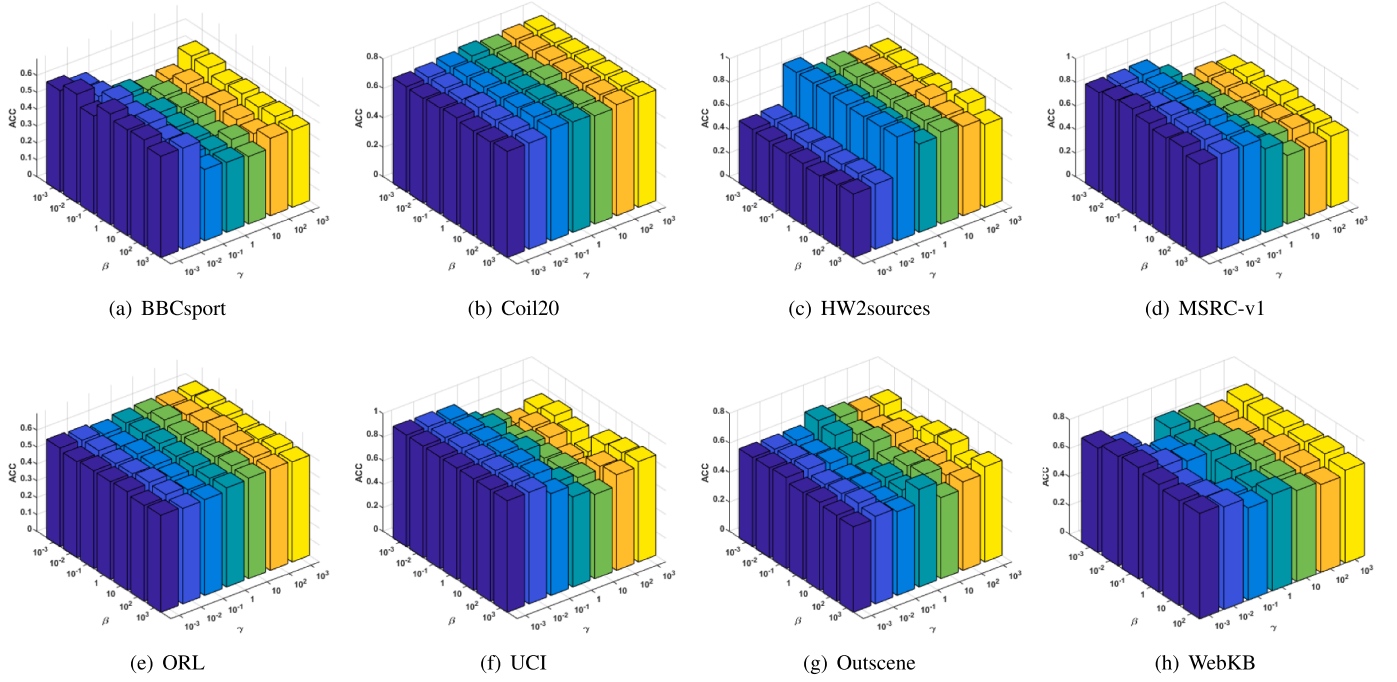
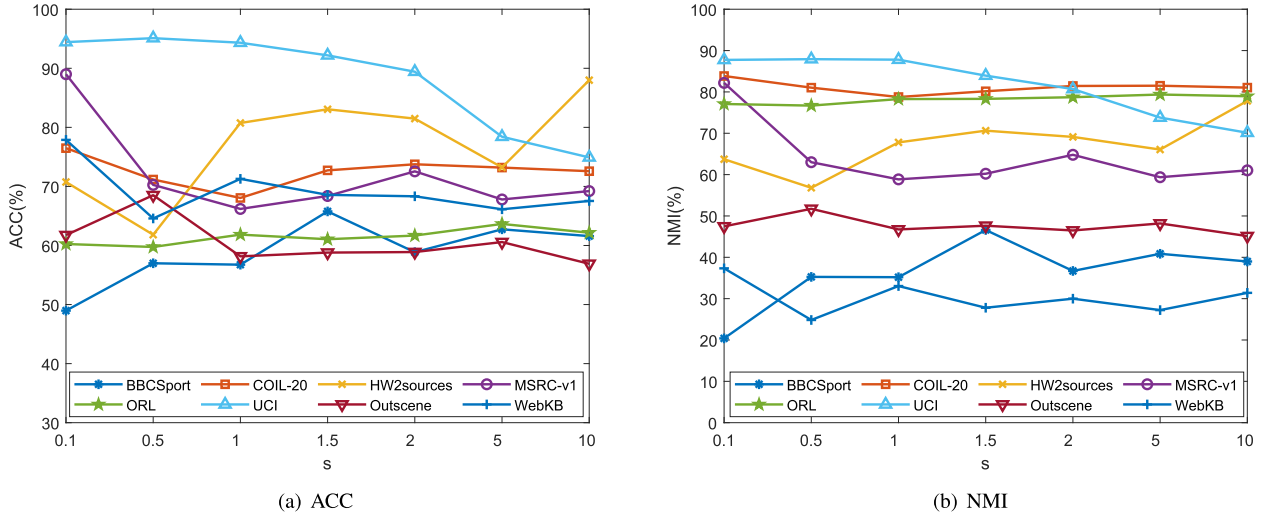
4.3. Experimental setup

To ensure a fair comparison, we set the hyperparameters of the methods strictly according to the parameter settings in the original literature. RSTR has three main hyperparameters: α , β , and γ . They are tuned within the range of $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$. For the view

Fig. 5. ACC results of RSTR w.r.t. α and β on all datasets.Fig. 6. ACC results of RSTR w.r.t. α and γ on all datasets.

weight parameter ω , it is tuned using a grid search strategy within the range $[1, 50]$. Meanwhile, the predetermined scale parameter s is tuned over the range $\{0.1, 0.5, 1, 1.5, 2, 5, 10\}$. For each dataset, all data matrices are normalized so that each sample vector has a unit ℓ_2 -norm. This operation removes scale bias among samples and ensures numerical stability during optimization. To evaluate the quality of the selected features, we perform clustering experiments following these steps. For multi-view data, the projection matrices W^v from all views are first concatenated along the feature dimension. Then, the ℓ_2 -norm of each row in the concatenated matrix is computed to obtain feature

importance scores, with each row corresponding to a feature. The features are ranked in descending order according to these scores, and the top k features are then selected to construct reduced datasets, with k chosen from $\{5\%, 10\%, 15\%, 20\%, 25\%\}$ of the total number of features. The clustering is performed using K-means on the reduced features with the number of clusters set to the ground-truth class count (c). To reduce the effect of random initialization, each experiment is repeated 40 times, and the clustering results are compared with the true labels. The final performance is reported as the average over these 40 runs. Two widely used metrics, accuracy (ACC) and normalized

Fig. 7. ACC results of RSTR w.r.t. β and γ on all datasets.Fig. 8. Sensitivity analysis of parameter s .

mutual information (NMI), are employed to evaluate the clustering quality, with higher values indicating better feature selection performance. At the same time, we recorded the runtime required for each method. It is worth noting that the reported runtime refers to the execution time of the feature selection process only, while the subsequent clustering step is excluded. Therefore, baseline methods that only perform clustering without feature selection are not included in the runtime comparison.

4.4. Experimental results and analysis

For a fair comparison, we set the number of selected features to 15 %, and Tables 3 and 4 show the clustering results for all methods under the optimal parameters. The optimal results are in bold, and the sub-optimal results are underlined. Figs. 2 and 3 illustrate the effects of

selecting varying numbers of features on the clustering performance. Table 5 shows the running time of different methods on each dataset. Note that “N/A” indicates that the runtime exceeded the four-hour limit under our experimental configuration (Intel Core i9-14900K CPU with 128 GB RAM). The following conclusions can be inferred from these tables and figures.

- In most cases, the RSTR method achieves either the optimal or the suboptimal performance. Compared with competitors, the improvement provided by RSTR is substantial. For instance, on the UCI dataset, our method outperforms the suboptimal approach by approximately 10.83% in ACC and 10.47% in NMI. Moreover, it also achieves satisfactory results on the image datasets Coil20 and Outscene, demonstrating the effectiveness of the proposed approach.

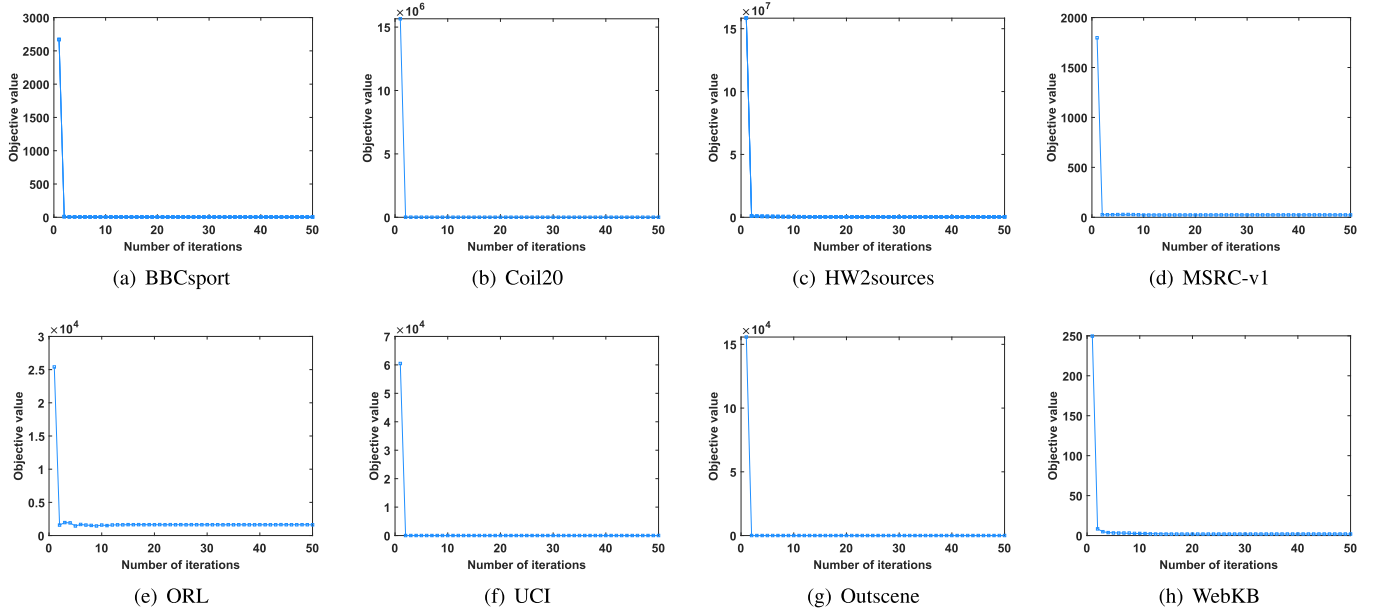


Fig. 9. Convergence curves of RSTR on all datasets.

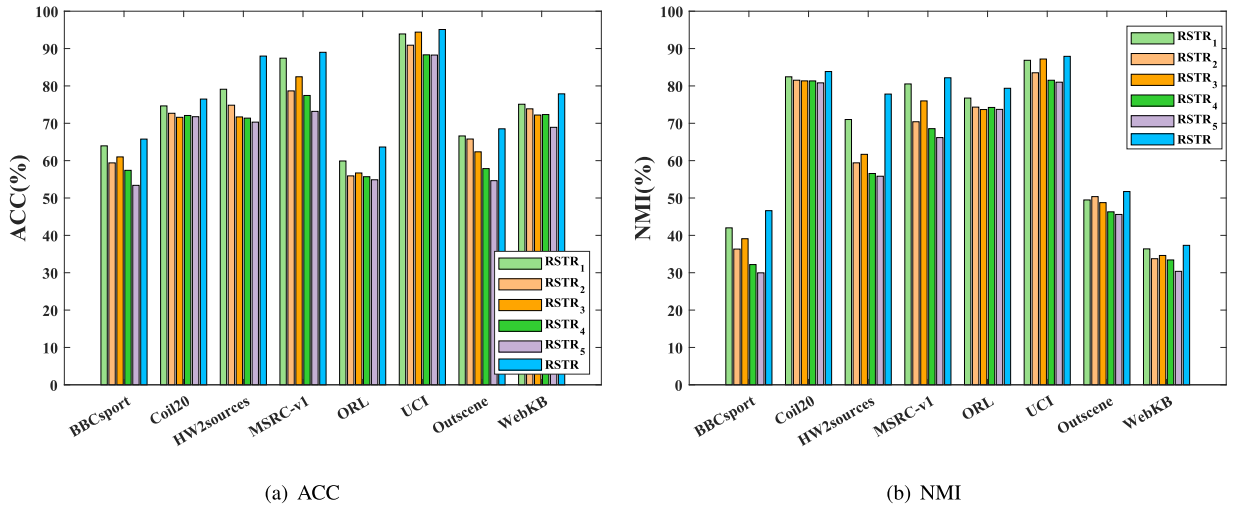


Fig. 10. ACC and NMI results of RSTR and five variants on eight datasets.

It is worth noting that increasing the proportion of selected features does not always lead to improved performance, indicating the presence of noisy and redundant features in the data. This observation underscores the necessity of performing feature selection to enhance the model performance.

- RSTR demonstrates superior performance compared with other self-representation models, such as GCDUFS. This improvement can be attributed to several factors. First, the separation of noise effectively removes its influence from the original data. Second, the application of the scaled simplex constraint on the clean representation allows for more flexible exploration of the relationships among data points. Third, the proposed rank constraint leverages the partition information of the data to ensure that the affinity matrix exhibits an optimal structure. Finally, stacking the high-quality affinity matrices into a weighted tensor further facilitates the capture of higher-order

relationships across multiple views.

- We observe that WLTl and JMVFG achieve the highest NMI on the ORL and WebKB datasets, respectively, while the RSTR model performs slightly worse on these datasets. Through analysis of these methods, we find that both WLTl and JMVFG leverage pseudo-labels to guide feature selection. This suggests that, for these datasets, using pseudo-label optimization can more effectively capture the underlying cluster structure, thereby leading to a higher NMI.
- From the perspective of computational time, RSTR requires more processing time due to the computation of $n \times n$ matrices. In addition, although GCDUFS also constructs $n \times n$ matrices, it concatenates the features of all views to explore cross-view consistency, which becomes particularly expensive on high-dimensional datasets. Therefore, our method achieves a favorable trade-off between performance and efficiency.

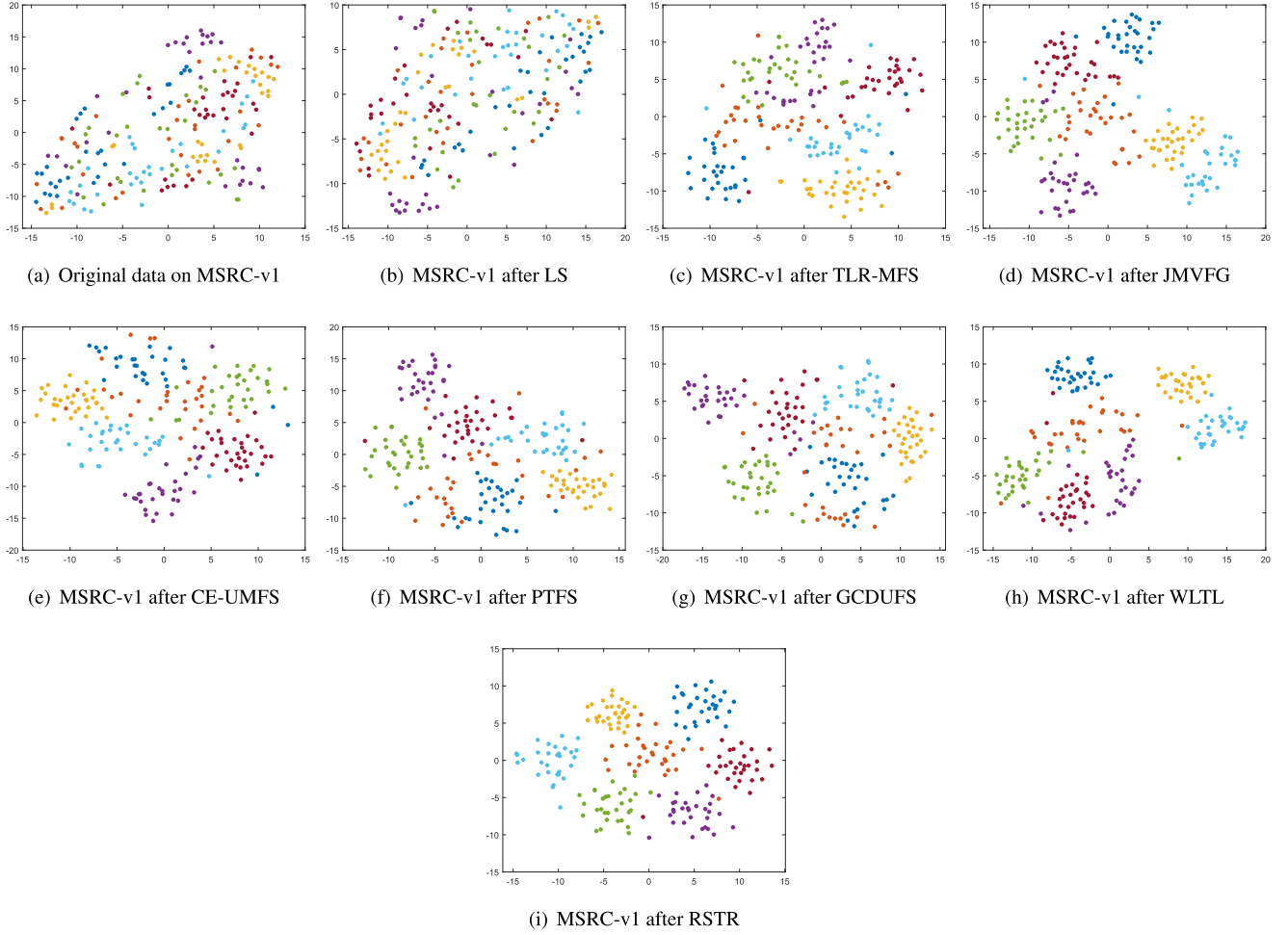


Fig. 11. Visualization of the clustering structures of nine methods on the MSRC-v1 dataset.

4.5. Parameter analysis

Due to the different characteristics of different datasets, different datasets will therefore correspond to different optimal parameter choices. The purpose of parameter analysis is to study the impact of different parameters on the performance of RSTR. Figs. 4 ~ 8 illustrate the sensitivity analysis of the RSTR method with regard to various parameters, including the view weight ω , the hyperparameters α , β , and γ , and the scaled simplex parameter s .

- Fig. 4 illustrates the impact of different view weight ω on ACC and NMI for the UCI and Coil20 datasets. The results indicate that ω plays a crucial role in improving the performance of RSTR.
- As shown in Figs. 5–7, it can be seen that when adjusting the values of α , β , and γ , RSTR performs relatively stably on most datasets, especially on the ORL and Coil20 datasets. The α regulates the sparse regularization term of the feature selection matrix. When α is in the range of $[10^{-1}, 10^1]$, RSTR achieves the best performance on most datasets. The β controls the term of the noise matrix. When β is in the range of $[10^{-1}, 10^1]$, RSTR performs best on most datasets, suggesting that β within this range effectively reduces the negative impact of noise on the model. The γ adjusts the impact of the rank constraint. It shows strong sensitivity on certain datasets. For example, on the HW2sources dataset, the values of γ need to be precisely tuned to achieve the best results, indicating that the rank constraint has a significant impact on some datasets.

- Fig. 8 demonstrates the effects of s on the model, where the other parameters have been set as optimal. It can be seen that changing the value of s has a large impact on all the datasets, which indicates that the scaled simplex constraint plays an important role in improving the performance of feature selection.

4.6. Convergence analysis

The convergence analysis of ALM-ADM is difficult for models with more than two components. Thus, the convergence analysis is conducted by plotting the convergence curves for RSTR across eight datasets. As shown in Fig. 9, the convergence curves for RSTR stabilize quickly on all datasets except ORL. The perturbations observed in the ORL dataset are mainly attributed to the inherent characteristics of the dataset itself and the initialization method used in the experimental setup. The results demonstrate that the proposed method can converge quickly.

4.7. Ablation studies

In this article, the self-representation matrix is decomposed into clean and noise components, which helps mitigate the influence of noise in the original data. The scaled simplex constraint is applied to the clean component to better capture the inherent relationships among the data points, while the rank constraint ensures a more optimal matrix structure. To investigate whether these strategies are practically helpful for the feature selection task, we design five variants of RSTR:

- **RSTR₁**: By removing the noise matrix term, i.e., setting the β term to 0, we get RSTR₁.
- **RSTR₂**: By removing the scaled simplex constraint term, i.e., setting the value of s to 1, we get RSTR₂.
- **RSTR₃**: By removing the rank constraint term, i.e., setting the γ term to 0, we get RSTR₃.
- **RSTR₄**: By removing the noise matrix term and the scaled simplex constraint, i.e., setting the β term to 0 and fixing the scaling factor s to 1, we obtain RSTR₄.
- **RSTR₅**: By removing the noise matrix term, the scaled simplex constraint, and the rank constraint, i.e., setting the β term to 0, fixing the scaling factor s to 1, and setting the γ term to 0, we obtain RSTR₅.

As demonstrated in Fig. 10 (located after the reference list), RSTR consistently yields optimal results. It is worth noting that RSTR₂ and RSTR₃ generally perform worse than RSTR₁ across most datasets, emphasising the importance of imposing both scaled simplex constraint and rank constraint on the self-representation matrix. The experimental results validate the effectiveness of the proposed feature selection strategies.

4.8. Visualization

To demonstrate the performance of RSTR, we use t-SNE (Van der Maaten & Hinton, 2008) to visualize the clustering structures of the original data and the data after feature selection. As shown in Fig. 11 (located after the reference list), RSTR exhibits a clear clustering structure on the MSRC-v1 datasets, highlighting its ability to improve the performance of the downstream tasks and demonstrating the effectiveness of the method (Algorithm 1).

5. Conclusion

In this article, we propose a novel MUFS method, namely, structure-preservation tensorized representation (RSTR). Specifically, the affinity matrix is decomposed into a clean component and a noise component, which effectively mitigates the influence of noise in the original data. To better capture the inherent structure of the data, a scaled simplex constraint is imposed on the clean affinity matrix, offering greater flexibility in uncovering latent relationships. In addition, a rank constraint enhances the structural quality of the affinity matrix. Finally, the weighted clean affinity matrices from each view are combined into a tensor to explore higher-order relationships between views. The experimental results demonstrate the proposed RSTR method clearly outperforms the state-of-the-arts.

For future work, we intend to investigate reducing the number of tuning hyperparameters to lower the computational overhead. Moreover, we plan to endow RSTR with the ability to cope with large-scale MUFS tasks to further improve its applicability in real-world scenarios.

CRedit authorship contribution statement

Chen Wang: Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation; **Peng Song**: Writing – review & editing, Supervision, Methodology, Funding acquisition; **Changjia Wang**: Writing – review & editing, Investigation, Data curation; **Beihua Yang**: Writing – review & editing; **Zhaowei Liu**: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Bertsekas, D. P. (2014). Constrained optimization and Lagrange multiplier methods. Academic press.
- Cai, B., Lu, G. F., Li, H., & Song, W. (2024). Tensorized scaled simplex representation for multi-view clustering. *IEEE Transactions on Multimedia*, 26, 6621–6631.
- Candès, E. J., Li, X., Ma, Y., & Wright, J. (2011). Robust principal component analysis? *Journal of the ACM (JACM)*, 58(3), 1–37.
- Cao, Z., & Xie, X. (2024a). Partition-level tensor learning-based multiview unsupervised feature selection. *IEEE Transactions on Neural Networks and Learning Systems*, (pp. 1–13).
- Cao, Z., & Xie, X. (2024b). Structure learning with consensus label information for multi-view unsupervised feature selection. *Expert Systems with Applications*, 238, 121893.
- Cao, Z., Xie, X., & Li, Y. (2024). Multi-view unsupervised feature selection with consensus partition and diverse graph. *Information Sciences*, 661, 120178.
- Chandrashekar, G., & Sahin, F. (2014). A survey on feature selection methods. *Computers & Electrical Engineering*, 40(1), 16–28.
- Chen, M. S., Wang, C. D., & Lai, J. H. (2022). Low-rank tensor based proximity learning for multi-view clustering. *IEEE Transactions on Knowledge and Data Engineering*, 35(5), 5076–5090.
- Chung, F. R. K. (1997). Spectral graph theory (vol. 92). American Mathematical Soc.
- Du, L., & Shen, Y. D. (2015). Unsupervised feature selection with adaptive structure learning. In *Proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 209–218).
- Duan, M., Song, P., Zhou, S., Cheng, Y., Mu, J., & Zheng, W. (2025). High-order correlation preserved multi-view unsupervised feature selection. *Engineering Applications of Artificial Intelligence*, 139, 109507.
- Fan, K. (1949). On a theorem of weyl concerning eigenvalues of linear transformations i. *Proceedings of the National Academy of Sciences*, 35(11), 652–655.
- Fang, S. G., Huang, D., Wang, C. D., & Tang, Y. (2023a). Joint multi-view unsupervised feature selection and graph learning. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 8(1), 16–31.
- Fang, U., Li, M., Li, J., Gao, L., Jia, T., & Zhang, Y. (2023b). A comprehensive survey on multi-view clustering. *IEEE Transactions on Knowledge and Data Engineering*, 35(12), 12350–12368.
- Gao, Q., Xia, W., Wan, Z., Xie, D., & Zhang, P. (2020). Tensor-SVD based graph learning for multi-view subspace clustering. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 3930–3937). (vol. 34).
- Guo, J., Sun, Y., Gao, J., Hu, Y., & Yin, B. (2020). Robust adaptive linear discriminant analysis with bidirectional reconstruction constraint. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 14(6), 1–20.
- Guo, J., Sun, Y., Gao, J., Hu, Y., & Yin, B. (2022). Multi-attribute subspace clustering via auto-weighted tensor nuclear norm minimization. *IEEE Transactions on Image Processing*, 31, 7191–7205.
- Han, Z., Zhang, C., Fu, H., & Zhou, J. T. (2022). Trusted multi-view classification with dynamic evidential fusion. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2), 2551–2566.
- He, X., Cai, D., & Niyogi, P. (2005). Laplacian score for feature selection. *Advances in Neural Information Processing Systems*, 18.
- Hu, R., Gan, J., Zhan, M., Li, L., & Wei, M. (2025). Unsupervised kernel-based multi-view feature selection with robust self-representation and binary hashing. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 17287–17294). (vol. 39).
- Huang, J., Nie, F., & Huang, H. (2015). A new simplex sparse learning model to measure data similarity for clustering. In *Proceedings of the 24th international conference on artificial intelligence* (pp. 3569–3575).
- Huang, P., Kong, Z., Xie, M., & Yang, X. (2023). Robust unsupervised feature selection via data relationship learning. *Pattern Recognition*, 142, 109676.
- Kilmer, M. E., Braman, K., Hao, N., & Hoover, R. C. (2013). Third-order tensors as operators on matrices: A theoretical and computational framework with applications in imaging. *SIAM Journal on Matrix Analysis and Applications*, 34(1), 148–172.
- Kilmer, M. E., & Martin, C. D. (2011). Factorization strategies for third-order tensors. *Linear Algebra and its Applications*, 435(3), 641–658.
- Li, W., Chen, H., Li, T., Wan, J., & Sang, B. (2022). Unsupervised feature selection via self-paced learning and low-redundant regularization. *Knowledge-Based Systems*, 240, 108150.
- Li, Z., Yang, Y., Liu, J., Zhou, X., & Lu, H. (2012). Unsupervised feature selection using nonnegative spectral analysis. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1026–1032). (vol. 26).
- Liang, S., Xu, Q., Zhu, P., Hu, Q., & Zhang, C. (2017). Unsupervised feature selection by manifold regularized self-representation. In *2017 IEEE international conference on image processing (ICIP)* (pp. 2398–2402). IEEE.
- Lin, Q., Men, M., Yang, L., & Zhong, P. (2022). A supervised multi-view feature selection method based on locally sparse regularization and block computing. *Information Sciences*, 582, 146–166.
- Liu, Z., & Song, P. (2024). Deep low-rank tensor embedding for multi-view subspace clustering. *Expert Systems with Applications*, 237, 121518.
- Van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9(11).

- Mohar, B., Alavi, Y., Chartrand, G., & Oellermann, O. (1991). The Laplacian spectrum of graphs. *Graph Theory, Combinatorics, and Applications*, 2(12), 871–898.
- Nie, F., Wang, X., & Huang, H. (2014). Clustering and projected clustering with adaptive neighbors. In *Proceedings of the 20th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 977–986).
- Nie, F., Wang, X., Jordan, M., & Huang, H. (2016). The constrained Laplacian rank algorithm for graph-based clustering. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1969–1976). (vol. 30).
- Patterson, S., & Teh, Y. W. (2013). Stochastic gradient Riemannian Langevin dynamics on the probability simplex. *Advances in Neural Information Processing Systems*, 26.
- Salau, A. O., & Jain, S. (2019). Feature extraction: A survey of the types, techniques, applications. In *2019 international conference on signal processing and communication (ICSC)* (pp. 158–164). IEEE.
- Shang, R., Chi, H., Li, Y., & Jiao, L. (2023). Adaptive graph regularization and self-expression for noise-aware feature selection. *Neurocomputing*, 535, 107–122.
- Song, P., Zhou, S., Mu, J., Duan, M., Yu, Y., & Zheng, W. (2024). Clean affinity matrix induced hyper-Laplacian regularization for unsupervised multi-view feature selection. *Information Sciences*, 682, 121276.
- Tang, C., Zheng, X., Zhang, W., Liu, X., Zhu, X., & Zhu, E. (2023). Unsupervised feature selection via multiple graph fusion and feature weight learning. *Science China Information Sciences*, 66(5), 152101.
- Van Engelen, J. E., & Hoos, H. H. (2020). A survey on semi-supervised learning. *Machine Learning*, 109(2), 373–440.
- Wang, C., & Song, P. (2025). Robust multi-view unsupervised feature selection via latent consensus structure learning. *Information Sciences*, 713, 122177.
- Wang, C., Song, P., Duan, M., Zhou, S., & Cheng, Y. (2025a). Low-rank tensor based smooth representation learning for multi-view unsupervised feature selection. *Knowledge-Based Systems*, 309, 112902.
- Wang, D., Wang, L., Chen, W., Wang, H., & Liang, C. (2025b). Unsupervised multi-view feature selection based on weighted low-rank tensor learning and its application in multi-omics datasets. *Engineering Applications of Artificial Intelligence*, 143, 110041.
- Wang, F., Zhu, L., Li, J., Chen, H., & Zhang, H. (2021). Unsupervised soft-label feature selection. *Knowledge-Based Systems*, 219, 106847.
- Wang, S., Nie, F., Wang, Z., Wang, R., & Li, X. (2023). Outliers robust unsupervised feature selection for structured sparse subspace. *IEEE Transactions on Knowledge and Data Engineering*, 36(3), 1234–1248.
- Xie, X., Sun, F., Qian, J., Guo, L., Zhang, R., Ye, X., & Wang, Z. (2023). Laplacian L_p norm least squares twin support vector machine. *Pattern Recognition*, 136, 109192.
- Xu, J., Yu, M., Shao, L., Zuo, W., Meng, D., Zhang, L., & Zhang, D. (2019). Scaled simplex representation for subspace clustering. *IEEE Transactions on Cybernetics*, 51(3), 1493–1505.
- Xu, S., Xie, X., & Cao, Z. (2025). Graph-regularized consensus learning and diversity representation for unsupervised multi-view feature selection. *Knowledge-Based Systems*, (p. 113043).
- Yang, X., Che, H., Leung, M.-F., & Wen, S. (2024). Self-paced regularized adaptive multi-view unsupervised feature selection. *Neural Networks*, 175, 106295.
- Yuan, H., Li, J., Liang, Y., & Tang, Y. Y. (2022). Multi-view unsupervised feature selection with tensor low-rank minimization. *Neurocomputing*, 487, 75–85.
- Zhang, Z., Ely, G., Aeron, S., Hao, N., & Kilmer, M. (2014). Novel methods for multilinear data completion and de-noising based on tensor-SVD. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 3842–3849).
- Zhou, P., Du, L., Li, X., Shen, Y. D., & Qian, Y. (2020). Unsupervised feature selection with adaptive multiple graph learning. *Pattern Recognition*, 105, 107375.
- Zhou, S., & Song, P. (2024). Consistency-exclusivity guided unsupervised multi-view feature selection. *Neurocomputing*, 569, 127119.